

VIKASH MADDI

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EDUCATION

Indian Institute of Technology Madras, Chennai, India

July 2018 – May 2022

Bachelor of Technology in Mechanical Engineering

8.31/10

PROFESSIONAL EXPERIENCE

VMock, Full Stack Developer

August 2022 – Present

Full-Stack Product Delivery

- Delivered **GenAI-powered** resume suggestions through end-to-end **React** UI and backend integration, giving users real-time personalized feedback.
- Built self-serve internal tooling and admin dashboards that automated onboarding and tenant provisioning, removing 30% of manual operational work.
- Designed and released a versioned **Flask** microservice with **REST API** that replaced a monolithic utility module, providing clear documentation for downstream teams.

LLM Agent Platform & Observability

- Built and operate a production **LLM** agent used by 5,000+ users, orchestrating tool calls with **Pydantic**-validated schemas as guardrails.
- Stood up **LLM** observability using **Arize Phoenix**, instrumenting span-level traces so failures can be attributed to model, prompt, tool, or harness.
- Implemented token, cost and latency tracking per session and added prompt-caching that cut inference cost for long-running sessions.

Reliability & Performance Engineering

- Improved workflow reliability with stateful context management and a two-tier retry strategy, cutting incorrect tool executions by 40%.
- Cut API response latency from **400ms** to **30ms** (13x) by adding indexes, **Redis** caching and refining service boundaries.
- Run event-driven workers on **AWS SQS** with explicit retry, timeout and idempotency semantics, ensuring durable processing.

Karomi Technologies, Deep Learning Intern

May 2021 – August 2021

- Trained Faster R-CNN, RetinaNet and EfficientDet models in **PyTorch** with semi-supervised learning, achieving mAP of 96% for tables and 95% for symbols.

PROJECTS

Context Management Proxy and MCP Server | *Python, MCP, Agent Infrastructure*

2026

- Identified cost growth in long-running coding-agent tasks and built a transparent HTTP proxy and **MCP** server that compresses agent context without client changes.
- Implemented relevance-based eviction and prefix-stable prompt caching, then measured cost per completed task before and after compression to prove the improvement.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Bash, C++, PHP, Java

Agentic AI Stack: LangGraph, MCP (authoring and hosting servers), Tool-Use Frameworks, Agent Orchestration, Guardrails, OpenAI/Anthropic APIs, Context Management, Prompt Caching, Claude Code, Cursor

Backend & APIs: FastAPI, Flask, Laravel, REST APIs, Microservices, asyncio, Pydantic, API Versioning

Frontend: React, Redux, Tailwind, HTML5, CSS3

Containers & Cloud: Kubernetes, Docker, AWS (S3, SQS, IAM basics), Cloudflare Workers, Linux, Deployment Automation, Rollback Strategy

AI Observability & Evaluation: Arize Phoenix, Span-Level Tracing, LLM-as-Judge Graders, Regression Suites, Token/Cost/Latency Tracking

SCHOLASTIC ACHIEVEMENTS

- Secured a global rank of 38 in the July Long Challenge on the CodeChef platform.
- Secured an all-India open category rank of 1308 in JEE-ADVANCED 2018 among 1.8 million.